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Environment: TryHackMe What is Networking? Room + Windows local network diagnostics

Focus Area: Network Fundamentals, ICMP Connectivity Testing & Operational Troubleshooting

Network Connectivity & ICMP Diagnostic Review

Task 1 What is Networking?

Networks are simply things connected. For example, your friendship circle: you are all connected because of similar interests, hobbies, skills and sorts.

Networks can be found in all walks of life:

- A city's public transportation system
- Infrastructure such as the national power grid for electricity
- Meeting and greeting your neighbours
- Postal systems for sending letters and parcels

But more specifically, in computing, networking is the same idea, just dispersed to technological devices. Take your phone as an example; the reason that you have it is to access things. We'll cover how these devices communicate with each other and the rules that follow.

In computing, a network can be formed by anywhere from 2 devices to billions. These devices include everything from your laptop and phone to security cameras, traffic lights and even farming!

Networks are integrated into our everyday life. Be it gathering data for the weather, delivering electricity to homes or even determining who has the right of way at a road. Because networks are so embedded in the modern-day, networking is an essential concept to grasp in cybersecurity.

Take the diagram below as an example, Alice, Bob and Jim have formed their network! We'll come onto this a bit later on.



Overview

A network connectivity and ICMP diagnostic review was conducted to reinforce foundational networking concepts used in IT Support, Monitoring, NOC, and Security Operations environments. The investigation focused on device identification, IP addressing, MAC addressing, private/public network concepts, and basic ICMP connectivity testing.

Room completed (100%)

An **IP address** is a set of numbers that are divided into four octets. The value of each octet will summarise to be the IP address of the device on the network. This number is calculated through a technique known as IP addressing & subnetting, but that is for another day. What's important to understand here is that IP addresses can change from device to device but cannot be active simultaneously more than once within the same network.

IP Addresses follow a set of standards known as protocols. These protocols are the backbone of networking and force many devices to communicate in the same language, which is something that we'll come onto another time. However, we should recall that devices can be on both a private and public network. Depending on where they are will determine what type of IP address they have: a public or private IP address.

A public address is used to identify the device on the Internet, whereas a private address is used to identify a device amongst other devices. Take the table & screenshot below as an example. Here we have two devices on a private network:

Device Name	IP Address	IP Address Type
DESKTOP-KJES7FD	192.168.1.77	Private
DESKTOP-KJES7FD	86.157.52.21	Public
CMNatic-PC	192.168.1.74	Private
CMNatic-PC	86.157.52.21	Public

DESKTOP-KJES7FD
5 GHz

IP address: 192.168.1.77 (DHCP)
MAC address: EC:5C:68:C3:7E:51

CMNatic-PC
5 GHz

IP address: 192.168.1.74 (DHCP)
MAC address: 50:3E:AA:E8:3B:64

Network Identification Review

IP addressing was reviewed to understand how devices are identified within a network. IPv4 addresses use four octets and allow systems to communicate within private networks or across public networks depending on the addressing context.

These two devices will be able to use their private IP addresses to communicate with each other. However, any data sent to the Internet from either of these devices will be identified by the same public IP address. Public IP addresses are given by your Internet Service Provider (or ISP) at a monthly fee (your bill!)

My Public IPv4 is: 86.157.52.21
Location: ENG GB
ISP: Secure Communications TLD

As more and more devices become connected, it is becoming increasingly harder to get a public address that isn't already in use. For example, Cisco, an industry giant in the world of networking, estimated that there would be approximately 50 billion devices connected on the Internet by the end of 2021. (Cisco, 2021). Enter IP address versions. So far, we have only discussed one version of the Internet Protocol addressing scheme known as IPv4, which uses a numbering system of 2^{32} IP addresses (4.29 billion) -- so you can see why there is such a shortage!

IPv6 is a new iteration of the Internet Protocol addressing scheme to help tackle this issue. Although it is seemingly more daunting, it boasts a few benefits:

- Supports up to 2^{128} of IP addresses (340 trillion-plus), resolving the issues faced with IPv4
- More efficient due to new methodologies

The screenshot below compares both an IPv6 and IPv4 address.

My IP Address is:
IPv6: **2a00:22c4:a531:c500:425f:cce6:c36b:f64d**
IPv4: **86.157.52.21**

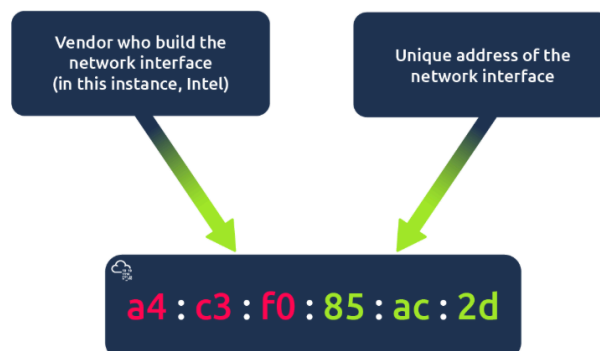
MAC Addresses

Private and Public Network Review

Private IP addresses were reviewed as identifiers used inside local networks, while public IP addresses were reviewed as identifiers used for communication across the Internet. This distinction is important for troubleshooting user connectivity issues and understanding network visibility.

MAC Addresses

Devices on a network will all have a physical network interface, which is a microchip board found on the device's motherboard. This network interface is assigned a unique address at the factory it was built at, called a **MAC (Media Access Control)** address. The MAC address is a **twelve-character** hexadecimal number (a base sixteen numbering system used in computing to represent numbers) split into two's and separated by a colon. These colons are considered separators. For example, `a4:c3:f0:85:ac:2d`. The first six characters represent the company that made the network interface, and the last six is a unique number.



However, an interesting thing with MAC addresses is that they can be faked or "spoofed" in a process known as spoofing. This spoofing occurs when a networked device pretends to identify as another using its MAC address. When this occurs, it can often break poorly implemented security designs that assume that devices talking on a network are trustworthy. Take the following scenario: A **firewall** is configured to allow any communication going to and from the MAC address of the administrator. If a device were to pretend or "spoof" this MAC address, the **firewall** would now think that it is receiving communication from the administrator when it isn't.

MAC Address Review

MAC addresses were reviewed as hardware-level identifiers assigned to network interfaces. Understanding MAC addresses is useful in troubleshooting local network access, device identification, and network access control scenarios.

Task 4 Ping (ICMP)

Ping is one of the most fundamental network tools available to us. Ping uses **ICMP** (Internet Control Message Protocol) packets to determine the performance of a connection between devices, for example, if the connection exists or is reliable. [View Site](#)

The time taken for ICMP packets travelling between devices is measured by ping, such as in the screenshot below. This measuring is done using ICMP's echo packet and then ICMP's echo reply from the target device.

Pings can be performed against devices on a network, such as your home network or resources like websites. This tool can be easily used and comes installed on Operating Systems (OSs) such as **Linux** and Windows. The syntax to do a simple ping is `ping IP address or website URL`. Let's see this in action in the screenshot below.

```
cmnatic@CMNatic-THM-LPTOP:~$ ping 192.168.1.254
PING 192.168.1.254 (192.168.1.254) 56(84) bytes of data:
64 bytes from 192.168.1.254: icmp_seq=1 ttl=63 time=2.18 ms
64 bytes from 192.168.1.254: icmp_seq=2 ttl=63 time=2.53 ms
64 bytes from 192.168.1.254: icmp_seq=3 ttl=63 time=2.15 ms
64 bytes from 192.168.1.254: icmp_seq=4 ttl=63 time=3.34 ms
64 bytes from 192.168.1.254: icmp_seq=5 ttl=63 time=10.3 ms
64 bytes from 192.168.1.254: icmp_seq=6 ttl=63 time=4.45 ms
^C
--- 192.168.1.254 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5008ms
rtt min/avg/max/mdev = 2.152/4.160/10.313/2.864 ms
cmnatic@CMNatic-THM-LPTOP:~$
```

Here we are pinging a device that has the private address of 192.168.1.254. Ping informs us that we have sent six ICMP packets, all of which were received with an average time of 4.16 milliseconds.

Now you are going to do the same thing to ping the address of "8.8.8.8" on the deployable website in this task. Pinging the correct address will reveal a flag to answer the following question below.

ICMP Connectivity Testing

Ping was reviewed as a basic network diagnostic tool that uses ICMP echo requests and replies to validate whether a target host is reachable and to measure basic response time.

```
user@thm:~$ ping -c 4 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=9.63 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=7.98 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=8.82 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=56 time=9.78 ms
--- 8.8.8.8 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 4006ms
rtt min/avg/max/mdev = 8.132/9.428/10.957/1.057 ms
Flag:
user@thm:~$ ping -c 4
```

Diagnostic Test

An ICMP connectivity test was performed against 8.8.8.8 to validate network reachability. Successful responses indicate that the target was reachable and that ICMP communication was functioning as expected in the lab environment.

```
Windows PowerShell
PS C:\Users\perdo> ipconfig

Configuración IP de Windows

Adaptador de Ethernet Ethernet:

    Estado de los medios. . . . . : medios desconectados
    Sufijo DNS específico para la conexión. . . :

Adaptador de Ethernet Ethernet 2:

    Sufijo DNS específico para la conexión. . . :
    Vínculo: dirección IPv6 local. . . : fe80::5c14:7e0c:45:e338%11
    Dirección IPv4. . . . . : 192.168.56.1
    Máscara de subred. . . . . : 255.255.255.0
    Puerta de enlace predeterminada. . . . . :

Adaptador de LAN inalámbrica Conexión de área local* 1:

    Estado de los medios. . . . . : medios desconectados
    Sufijo DNS específico para la conexión. . . :

Adaptador de LAN inalámbrica Conexión de área local* 2:

    Estado de los medios. . . . . : medios desconectados
    Sufijo DNS específico para la conexión. . . :

Adaptador de LAN inalámbrica Wi-Fi:
```

Windows Network Configuration Review

The Windows ipconfig command was used to review local network configuration, including IPv4 addressing, adapter status, subnet mask information, and default gateway configuration. This process represents a common first step in operational network troubleshooting and connectivity diagnostics.

```
Windows PowerShell
PS C:\Users\perdo> ipconfig /all

Configuración IP de Windows

Nombre de host. . . . . : DESKTOP-12J8DBM
Sufijo DNS principal. . . . . :
Tipo de nodo. . . . . : mixto
Enrutamiento IP habilitado. . . : no
Proxy WINS habilitado. . . . . : no

Adaptador de Ethernet Ethernet:

    Estado de los medios. . . . . : medios desconectados
    Sufijo DNS específico para la conexión. . . :
    Descripción. . . . . : Realtek PCIe GbE Family Controller
    Dirección física. . . . . : 4C-D7-17-3C-68-11
    DHCP habilitado. . . . . : sí
    Configuración automática habilitada. . . : sí

Adaptador de Ethernet Ethernet 2:

    Sufijo DNS específico para la conexión. . . :
    Descripción. . . . . : VirtualBox Host-Only Ethernet Adapter
    Dirección física. . . . . : 0A-00-27-00-00-0B
    DHCP habilitado. . . . . : no
    Configuración automática habilitada. . . : sí
    Vínculo: dirección IPv6 local. . . : fe80::5c14:7e0c:45:e338%11(Preferido)
    Dirección IPv4. . . . . : 192.168.56.1(Preferido)
    Máscara de subred. . . . . : 255.255.255.0
    Puerta de enlace predeterminada. . . . . :
```

Reviewed local network adapter configuration using ipconfig /all to verify IPv4 addressing, DNS configuration, gateway assignment, and adapter operational status.

```

Paleta de enlaces predeterminada : . . . . .
PS C:\Users\perdo> ping 8.8.8.8

Haciendo ping a 8.8.8.8 con 32 bytes de datos:
Respuesta desde 8.8.8.8: bytes=32 tiempo=23ms TTL=119
Respuesta desde 8.8.8.8: bytes=32 tiempo=21ms TTL=119
Respuesta desde 8.8.8.8: bytes=32 tiempo=22ms TTL=119
Respuesta desde 8.8.8.8: bytes=32 tiempo=23ms TTL=119

Estadísticas de ping para 8.8.8.8:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
    Tiempos aproximados de ida y vuelta en milisegundos:
        Mínimo = 21ms, Máximo = 23ms, Media = 22ms

```

Local Connectivity Validation

An ICMP connectivity test was performed from a Windows endpoint to 8.8.8.8 using the ping utility. Successful echo replies confirmed outbound network connectivity and demonstrated how ICMP testing can be used as an initial diagnostic step during network troubleshooting and connectivity validation procedures.

```

Mínimo = 21ms, Máximo = 23ms, Media = 22ms
PS C:\Users\perdo> nslookup google.com
Servidor:  csp1.zte.com.cn
Address:  fe80::1

Respuesta no autoritativa:
Nombre:  google.com
Addresses:  2607:f8b0:4008:808::200e
           172.217.165.206

```

DNS Resolution Validation

A DNS resolution test was performed using nslookup to verify successful domain name resolution from the local endpoint. The test confirmed that the configured DNS service was able to resolve external domains into valid IP addresses, demonstrating proper outbound DNS functionality.

```

PS C:\Users\perdo> tracert google.com

Traza a la dirección google.com [172.217.165.206]
sobre un máximo de 30 saltos:

 1  *          *          *          Tiempo de espera agotado para esta solicitud.
 2  8 ms       3 ms       3 ms       172.16.255.2 [172.16.255.2]
 3  5 ms       6 ms       9 ms       38.255.80.1
 4  6 ms       6 ms       5 ms       10.129.89.9 [10.129.89.9]
 5  22 ms      24 ms      22 ms      142.251.201.40
 6  21 ms      23 ms      26 ms      192.178.109.89
 7  22 ms      27 ms      22 ms      142.250.239.73
 8  *          *          26 ms      tzmiaa-ac-in-f14.1e100.net [172.217.165.206]

Traza completa.
PS C:\Users\perdo>

```

Route Tracing Review

A route tracing test was performed using tracert to observe the network path between the local endpoint and the external destination. This type of diagnostic test can help identify potential connectivity delays, routing issues, or packet loss across intermediary network hops during troubleshooting procedures.

Findings

- IP addressing is essential for identifying devices and troubleshooting connectivity.
- Private and public IP address concepts help explain internal versus external network communication.
- MAC addresses support local device identification and network access control concepts.
- ICMP ping testing can validate basic reachability between systems.
- Windows command-line tools such as ipconfig, ping, nslookup, and tracert support operational troubleshooting workflows.
- Network diagnostics must be interpreted in context because firewalls, DNS settings, or network restrictions may affect results.

Operational Recommendations

- Start connectivity troubleshooting by reviewing local IP configuration.
- Validate whether the issue is local, DNS-related, or external by testing both IP addresses and domain names.
- Use ping to confirm basic reachability when ICMP is allowed.
- Use nslookup to validate DNS resolution behavior.
- Use tracert to review possible routing delays or path-related issues.
- Document test results clearly with timestamps, observed behavior, and troubleshooting context.

Key Takeaways

This review reinforced the importance of basic networking diagnostics in IT Support, Monitoring, NOC, and Security Operations environments. Understanding IP addressing, MAC addressing, ICMP testing, DNS resolution, and route tracing helps analysts approach connectivity issues through a structured troubleshooting process.